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The accuracy of artificial and natural light measurements by actigraphs

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Abstract

Actigraphs are the reference standard for measuring light exposure in human non-laboratory experiments due to their portability and long battery lives. However, actigraphs typically have a limited illuminance operating range not representative of real-world conditions, and for many actigraphs, the accuracy of their light measurement has not been verified independently. We assessed the illuminances recorded by Activinsights GENEActiv Original and Philips Actiwatch 2 actigraphs in comparison to a calibrated, laboratory-standard photometer, under both artificial light-emitting diode (LED) and natural sunlight illuminations that might be encountered by a person under real-world conditions. We show that in response to ~20,000 lux white LED light, the GENEActiv and Actiwatch 2 underestimate illuminance by recording 50% and 25% of the true value, respectively. Under ~30,000 lux sunlight, the GENEActiv readily saturates whereas the Actiwatch 2 reports ~46% of the true illuminance. These underestimations are highly linear and we provide correction factors to estimate the illuminance levels of the ambient environment measured by the actigraphs. We also evaluate the application of neutral density filters for extending the operating range of both devices in natural sunlight illuminations (as high as 30,000 lux during our measurements) and demonstrate that this may be a viable approach for increasing the operating range of the Actiwatch 2 but not the GENEActiv. We conclude that both actigraphs provide good performance in monitoring the temporal patterning of light, whereas the absolute illuminance values require correction to accurately evaluate the effects of light intensity on human health and behaviours.

KEYWORDS

illuminance, LED, light exposure, photometer, sunlight

1 | INTRODUCTION

Photoentrainment to the pattern of light and dark influences a variety of human behavioural processes, including circadian rhythms, sleep, alertness and mood (Czeisler et al., 1986; Dawson & Campbell, 1991; Golden et al., 2005; LeGates, Fernandez, & Hattar, 2014; Rüger, Gordijn, Beersma, Vries, & Daan, 2006). Photoreceptors in the eye that transduce environmental light information to brain areas for mediating these processes (Altimus et al., 2010; Bellingham & Foster, 2002; Chen, Badea, & Hattar, 2011; Rollag, Berson, & Provencio, 2003) are affected in disease (Adhikari, Zele, Thomas, & Feigl, 2016; Dumpala, Zele, & Feigl, 2019; Joyce, Feigl, Kerr, Roeder, & Zele, 2018; Maynard, Zele, Kwan, & Feigl, 2017) and the magnitude and timing of light exposure are associated with the circadian rhythm phase (Duffy, Kronauer, & Czeisler, 1996), sleep quality (Ancoli-Israel et al., 2003; Phipps-Nelson, Redman, Dijk, & Rajaratnam, 2003) and mood (Rüger et al., 2006). The efficacy of light therapy in treating disorders, including

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sleep abnormalities and depression (Campbell et al., 1995; Golden et al., 2005; Mishima et al., 1994; Montgomery & Dennis, 2002; Rosenthal et al.&&, 1984; Terman & Terman, 2014), underscores the importance of understanding and optimizing light exposures for human well-being and health.

Although calibrated research-grade photometers and radiometers accurately quantify light levels, they are not practical for measuring the light exposure of a freely moving individual over time. The standard instrument used for this purpose is a wrist-worn actigraphy device (Bernhofer, Higgins, Daly, Burant, & Hornick, 2014; Flynn et al., 2014; Grandner, Kripke, & Langer, 2006; Higgins, Hornick, & Figueiro, 2010; Higgins, Winkelman, Lipson, Guo, & Rodgers, 2007; Liu et al., 2005; Shochat, Martin, Marler, & Ancoli-Israel, 2000). However, the accuracy of many of these devices in measuring environmental light exposure has not been verified independently, and those devices that have been independently measured show inaccuracies. For example, compared to a calibrated Extech Light Meter, the Sleepwatch-L device underestimates the indoor illuminance (11:00–17:00 hours; both artificial and natural lights) during daylight hours by up to 64% (Higgins et al., 2007). To further verify the use of actigraphs in measuring light exposure, here we determine the accuracy of environmental light-emitting diode (LED) light and natural sunlight measurements reported by Activinsights GENEActiv Original and Philips Actiwatch 2 devices.

2 | METHODS

2.1 | Actigraphs

Two commercially available actigraph devices were tested in this study, the GENEActiv Original (n = 6; Activinsights) and Actiwatch 2 (n = 6; Philips Respironics). The GENEActiv device has a stated operating range of 5–3,000 lux, 5 lux resolution, an accuracy of ±10% at 1,000 lux and a wavelength range of 400–1,100 nm (ActivInsights, 2015; Figure 1a). The Actiwatch 2 device has a stated operating range of 5–100,000 lux spanning 400–900 nm (other parameters not stated; Philips Respironics, 2008). Both devices include visible and partial infrared portions of the electromagnetic spectrum, and record illuminance rather than the spectral power distribution of the light.

2.2 | Protocol

The LED and sunlight illuminances were continuously measured with the GENEActiv and Actiwatch 2 devices as well as a calibrated ILT1700 photometer (International Light Technologies). For the artificial light measurements (Experiment 1), an enclosed light exposure cabinet was constructed with a high-reflectance white surface. The light exposure cabinet comprised a dimmable 4,000 K white light LED panel (Lumex NovaBlade LL9PX66N) typical of artificial lighting conditions in an office environment. The spectral power distribution (SPD) of the LED light was measured in the plane of the actigraph devices with a high-resolution spectroradiometer (Stellarnet Inc. EPP2000; Figure 1a). The luminaire used a Samsung LM 561 B-5630 LED chip on-board module controlled with a pulse width modulated (PWM) signal at 400 Hz, higher than the upper limit of human flicker perception (Smith, Pokorny, Lee, & Dacey, 2008; Zele & Vingrys, 2005). The LED panel had a 120° beam angle and high colour rendering index (>90) and was diffused by a flat cosine panel. The LED output was controlled using the DALI protocol with Dynet Software (Smartscape Automation). Measurements were conducted in a 6-hr protocol, where the light ranged from 4.39 to 20,600 lux and featured gradual increases and decreases in illuminance, periods of darkness and rapid stepwise changes in illuminance (Figure 1b).

In Experiment 2, we assessed the performance of the actigraphs under natural sunlight. The actigraphs and photometer sensor were placed on a level, unshaded surface on the roof of a building in Brisbane, Australia, on a bright sunny morning in

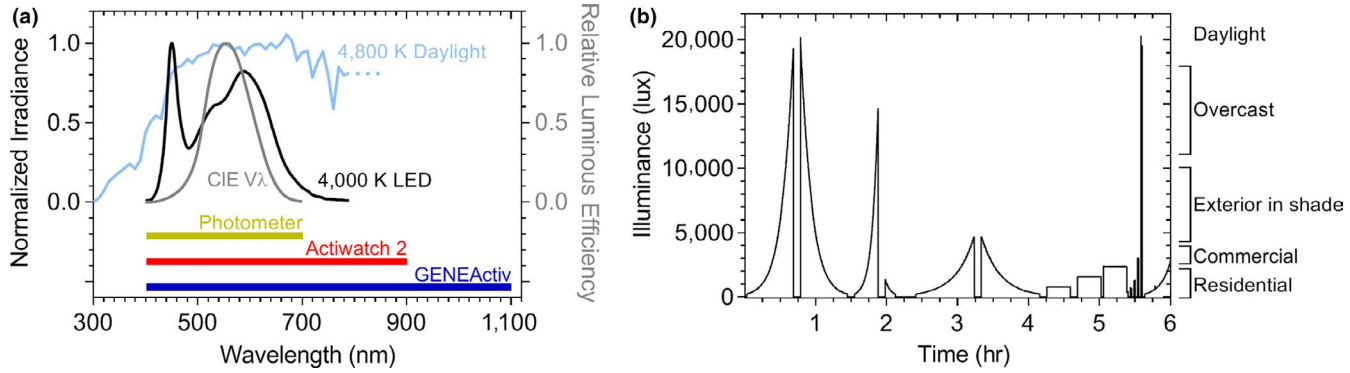


FIGURE 1 Light-emitting diode (LED) protocol. (a) Spectral power distribution of the LED light (black line) at a correlated colour temperature (CCT) of 4,000 K normalized to the peak at 453 nm and that of typical daylight (light blue line) at a CCT of 4,800 K normalized to the power at 555 nm (reproduced from Judd et al. (1964)); note that the daylight spectrum extends to shorter and longer wavelengths. The ILT1700 photometer's operating range (400–700 nm; yellow horizontal line) is defined by the International Commission on Illumination (CIE) photopic luminous efficiency function (Vλ; grey line; normalized to the peak at 555 nm), whereas the operating range for Actiwatch 2 and GENEActiv is 400–900 nm (red line) and 400–1,100 nm (blue line), respectively. (b) Stimulus protocol included slow ramps on and off, short and rapid stepwise increments, and periods of darkness

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December (summer) for 1 hr. To determine whether the operating ranges of the devices could be extended, we applied flexible, lightweight 0.3, 0.6 and 1.2 neutral density (ND) filters (LEE Filters) to the actigraphs photosensor. For all recordings, the actigraph wristbands were removed and the devices arranged at even intervals in a circle around the ILT1700 photometer sensor. The sampling frequencies of each device were set to 60 Hz and then down-sampled into 1-s epochs by each manufacturer's software. The photometer was set to an integration time of 500 ms and a measurement was taken every 500 ms, before averaging. For analyses, the data in Experiment 1 were averaged and in Experiments 1 and 2 the data were further down-sampled into 1-min epochs using MATLAB (R2016a; Mathworks). Post-hoc analysis revealed that the intra-device coefficients of variation (CoV, *SD*/mean) were <4% for both the GENEActiv and Actiwatch 2 devices.

3 | RESULTS

Under broadband LED illumination, compared to the photometer, both actigraph devices (GENEActiv, $n = 6$; Actiwatch 2, $n = 6$) underestimate the illuminance but show the same temporal light profile (Figure 2a). Both devices show systematic underestimation of the peak 20,800 lux illuminance reported by the photometer; the GENEActiv peaks at 9,527 lux, whereas the Actiwatch 2 peaks at 5,755 lux (Figure 2a), 46% and 28% of the photometer value, respectively (Figure 2b,d), with the underestimation independent of true illuminance. Although the devices underestimate the true illuminance, the relationships between photometer and actigraph outputs are highly linear (Figure 2c), including both within and above the 5–3,000 lux operating range of the GENEActiv device. The magnitude of underestimation by the actigraphs was most variable at illuminances <2,500 lux (Figure 2d).

We next evaluated the performance of the devices in measuring the illuminance of sunlight. As sunlight illuminance can readily exceed the stated operating range of both devices (upper limit: 3,000 lux for GENEActiv and 100,000 lux for Actiwatch 2), flexible neutral density (ND) filters were attached to individual devices to determine whether the operating range could be extended (0.3, 0.6, 1.2 ND filters and no filter [0 ND]; $n = 4$ GENEActiv and $n = 4$ Actiwatch 2). We show that under sunlight, without ND filters, both devices underestimate the true illuminance in comparison to the calibrated photometer (Figure 3a,b). The GENEActiv appears to saturate under these

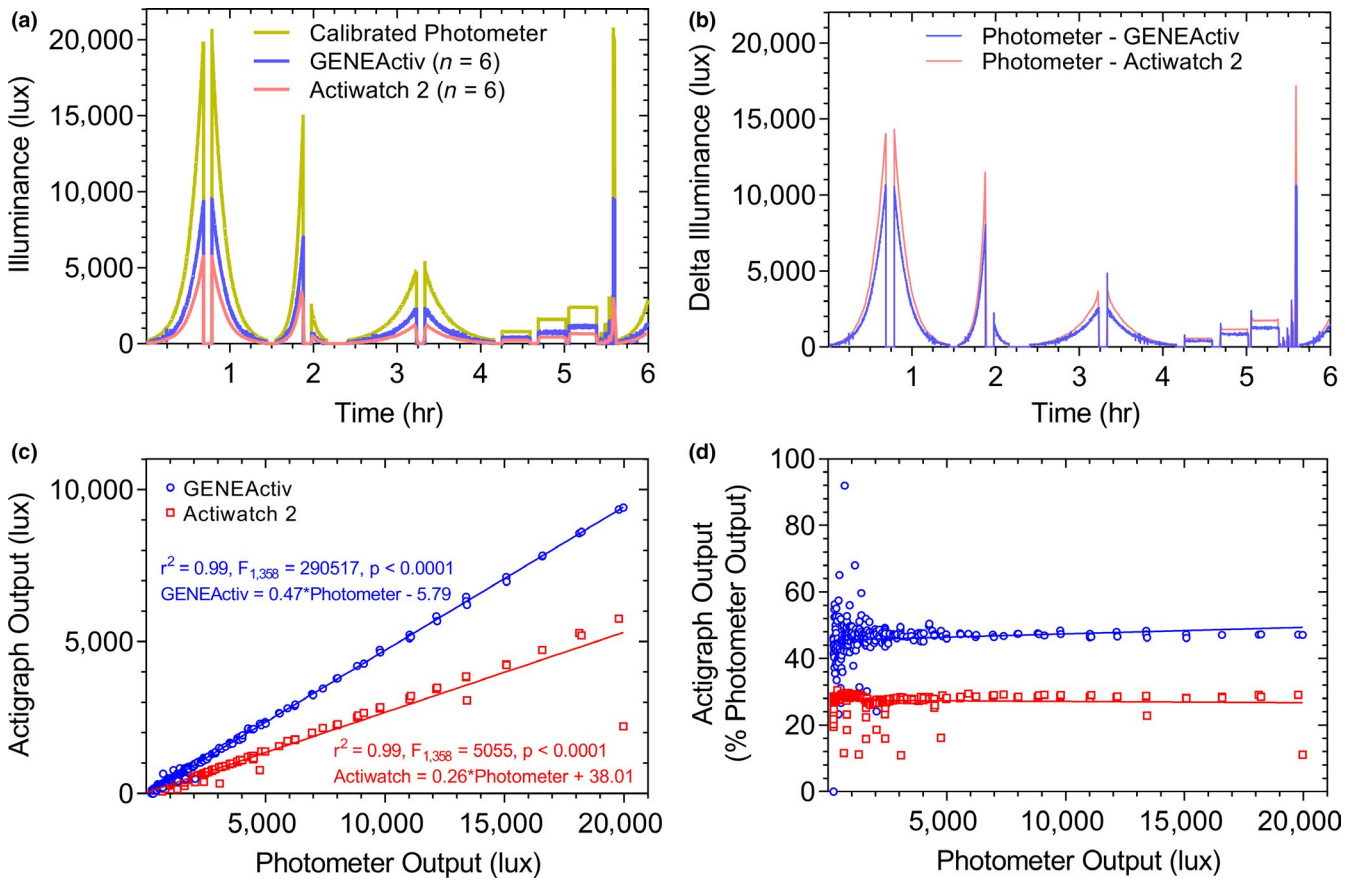


FIGURE 2 Light-emitting diode (LED) light measurements by actigraphs and calibrated photometer. (a) The averaged data from both actigraph devices show underestimation of the true illuminance as measured by the photometer, with the Actiwatch 2 being the most inaccurate. (b) Underestimates by the actigraphs increase as the true illuminance increases. The relationships between photometer illuminance and actigraph illuminances are linear (c) and demonstrate the highest variability, <2,500 lux, for both devices (d); the solid line indicates the best fitting linear regression

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conditions, reporting ~18,000 lux at peak, whereas the Actiwatch 2 does not. For the ND-attenuated devices, reported illuminance decreases as ND value increases as expected; however, this effect is non-systematic for the GENEActiv devices (*c.f.* 0.3 versus 0.6 ND tracings; Figure 3c). When the data are normalized to maximum, the tracings demonstrate poor concordance for the GENEActiv devices, indicative of saturation (Figure 3c), but good concordance for the Actiwatch 2 devices (Figure 3d). Response variability is higher under sunlight conditions for the Actiwatch 2 (Figure 3e) compared to LED light (Figure 2c). Actigraph outputs show a positive linear relationship with photometer outputs (Figure 3e,f). The Actiwatch 2 systematically underestimates illuminance, whereas the GENEActiv transitions from

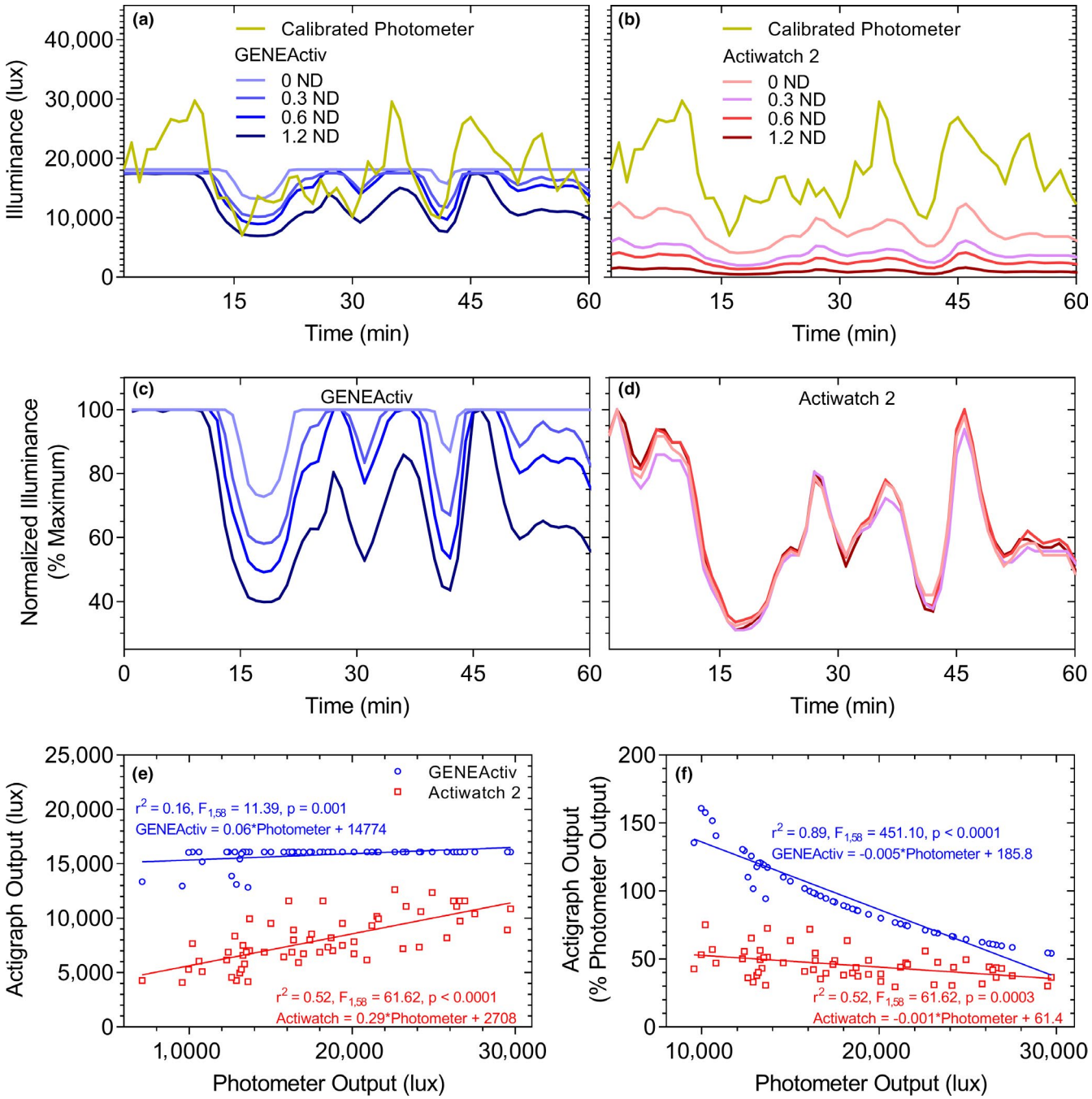


FIGURE 3 Sunlight measurements by actigraphs and calibrated photometer. Both actigraph devices underestimate the true illuminance as measured by the photometer; the GENEActivs saturate (a), whereas the Actiwatch 2s do not (b). Application of neutral density (ND) filters lowers the reported illuminance (a, b), although non-systematically for the GENEActivs. Normalizing these data to peak (c, d), GENEActiv tracings do not overlap, demonstrating that this method cannot counteract saturation of these illuminances, but Actiwatch 2 tracings overlap, suggesting that ND filters can be used to extend the operating range of this device. Plotting unfiltered actigraph irradiance against photometer irradiance (e) demonstrates the variability of underestimates, which are systematic for the Actiwatch 2s but not for the GeneActivs (f); the solid line indicates the best fitting linear regression

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overestimating to underestimating as illuminance increases (Figure 3f). The actigraph output expressed as a percentage of the corresponding photometer output decreases with increasing illuminance (Figure 3f), indicating that the magnitude of illuminance underestimation by the actigraphs increases with increasing light level. GENEActiv saturation at light levels >18,000 lux manifests in the regression, but the statistically significant regression is driven by the data <18,000 lux.

4 | DISCUSSION

This study evaluates the light-measurement performance of two commercially available actigraphs, the Activinsights GENEActiv and the Philips Actiwatch 2, under artificial (LED) illumination and sunlight. The data demonstrate that although both devices have good performance in monitoring the temporal patterning of light, they have poor illuminance-sensing accuracy under moderately intense artificial and natural sunlight conditions as might be encountered in the real world (Figures 2 and 3). In the absence of rigorous manufacturer-provided calibrations, our linear regression models provide a tool to scale the actigraph output for accurate representation of environmental illumination, including artificial as well as natural lights as measured by the laboratory standard photometer (see Data S1 and S2 for the toolboxes to correct GENEActiv and Actiwatch 2 outputs).

Across both artificial light and sunlight conditions, illuminance is typically underestimated by the actigraph devices but differed depending on the light level and source. For LED lighting, these underestimates are systematic and linear, particularly for higher intensities (Figure 2). Under sunlight, the estimates are more variable, with the GENEActiv transitioning from overestimating light intensity to underestimating once saturation occurred (Figure 3a,f). Sunlight contains the infrared (IR) spectrum, which, however, is negligible for our LED lights (0.6% of the total spectral power distribution; Figure 1a). The overestimation of sunlight illuminance may be due to the GENEActiv sensitivity (400–1,100 nm), which ranges into the infrared region, well beyond the human luminous efficiency functions (400–700 nm) used by the ILT1700 photometer to quantify illuminance (lux). Specifying the measurement of ambient illuminance in a spectrum broader than human vision can result in the intrusion of undesired radiometric components. Our analysis suggests that due to the highly linear relationship between device and photometer measurements, a conversion factor can be applied to estimate actual illuminance from that reported by the devices. This holds for LED lighting for both the GENEActiv and Actiwatch 2 (Figure 2c) and for sunlight for the Actiwatch 2 only (Figure 3e), but not for the GENEActiv under sunlight as it readily saturates under the conditions tested. The application of neutral density filters may be one cost-effective method to extend the operating range of these devices to include high light levels that may be encountered in daily living. Flexible neutral density filters ranging from 0.3 to 1.2 ND maintain the fidelity of the (filtered) light intensities recorded by the Actiwatch 2 (Figure 3b), but are ineffective for the GENEActiv due to continued saturation (Figure 3a). More generally, the data demonstrate that the absolute illuminance values reported by the actigraphs are not accurate at illuminances both within and outside of their stated operating ranges, either under artificial LED light or natural sunlight. This finding extends the growing body of evidence demonstrating significant inaccuracies in measuring light intensity with actigraphs, including substantial inter-device variation in light sensing between units from the same manufacturer (Cao & Barrionuevo, 2015; Price, Khazova, & O'Hagan, 2012).

Wrist-worn light-sensing devices have the potential to monitor naturalistic behaviours, including prolonged light exposures over days or weeks, in situations where traditional laboratory-based light measurement devices are impractical. Current-generation iterations of the GENEActiv and Actiwatch 2 can provide information on the temporal patterning of light, but they do not accurately report illuminance under either artificial or natural lighting conditions, both within and without their stated operating ranges. We encourage actigraph users to apply these correction factors to more accurately quantify ambient light levels. It would be advantageous if actigraphs included high-fidelity logging of not just the temporal patterning, but also the illuminance patterning of light across ecologically valid ranges, in both artificial and natural light. Increasing the accuracy and precision of these wearable devices is required to quantify, under naturalistic conditions, how the irradiance, patterning and spectral qualities of light are sensed by the image and non-image forming visual pathways to drive human behaviour.

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CONFLICT OF INTEREST

No conflicts of interest declared.

AUTHOR CONTRIBUTIONS

DSJ, AJZ, BF and PA designed research. AJZ and PA performed data collection. DSJ and PA performed data analysis and interpretation. DSJ and PA wrote the manuscript. DSJ, AJZ, BF and PA edited the manuscript.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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